



**Sandhya Rai**

Roll No.: 2206085

Computer Science and Engineering

B.Tech

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## EDUCATION

### •National Institute Of Technology, Patna

B.Tech in Computer Science and Engineering

2022-2026

6.79/CGPA: ongoing

## EXPERIENCE

### •Prodigy Infotech

Full-Stack Web Development

16/08/2024

Online

### •VaultofCodes

Web Development Intern

01/06/2025 – 01/07/2025

Online

- Successfully completed a 1-month internship, gaining hands-on web development experience.
- Contributed to project tasks and development cycles.

## PERSONAL PROJECTS

### SumiTask – Smart Revision & Task Scheduler

React.js, Electron, Node.js, Express.js, SQLite

- Developed a full-featured Desktop productivity app for students focused on spaced revision, daily planning, and task automation.
- Integrated Google OAuth login, dynamic task calendar, customizable spaced repetition intervals, and real-time task management.
- Enabled daily PDF summary generation using pdfkit and delivery via WhatsApp using Twilio Cloud API.
- Added Blank Page Revision: users upload study materials, later upload recall pages for active recall and future AI comparison.
- Structured the system for future AI-powered features like smart quiz generation, revision accuracy scoring, and voice assistant.
- Designed with modular architecture, scalable APIs and local storage.

### User Allocation in Edge Computing (QCSO) | Research Project

Python, Quantum-Inspired Differential Evolution, Metaheuristics

- Conducted in-depth data analysis on large datasets (5000 users) to identify performance bottlenecks in edge computing user allocation.
- Performed rigorous benchmarking of a novel QCSO algorithm against competing solutions (DE, PSO, CSO) to validate performance improvements.
- Demonstrated strong problem-solving skills by designing a solution that optimized for four key business metrics: energy consumption, load balancing, delay, and user allocation rate.
- Authored a detailed specification (academic paper) to communicate complex technical concepts and present data-driven conclusions, demonstrating excellent written communication skills.

## TECHNICAL SKILLS AND INTERESTS

**Languages:** C++, C, HTML, CSS, JavaScript, Python

**Developer Tools:** VS Code, DEV C++, Git, GitHub, Chrome DevTools, Power BI, Excel, PowerPoint

**Frameworks:** ReactJS, NodeJS

**Cloud/Databases:** MySQL, MongoDB

**Soft Skills:** Hardworking, Good Communication Skill, book reading

**Coursework:** Data Structure and Algorithms, Operating System, DBMS

**Areas of Interest:** Software Development, fullstack web development.

## POSITIONS OF RESPONSIBILITY

### •NSS Volunteer National Service Scheme

2023 – Present

## ACHIEVEMENTS

•**JEE Mains RankHolder** Obtained an All India Rank of 25278 among 15+ Lacs of Candidates.

•**Candidate - TCS iON Assessment** Successfully cleared the TCS iON assessment with a High Performance category

•**Adobe India Hackathon** Participated in Online MCQ Assessment + Coding as part of Team Matrix. 2025